Pharmacy Management System Database Normalization

Table-wise Normalization Process

1. Users Table

Functional Dependencies (FDs):

- user_id → username, password_hash, role
- username → user_id, password_hash, role *(if username is unique)*

Checking Normal Forms:

UNF to 1NF: Already in 1NF (all attributes are atomic).
1NF to 2NF: No partial dependencies (all non-key attributes depend on user_id).
2NF to 3NF: No transitive dependencies.
3NF to BCNF: Every determinant is a candidate key.

Final Normal Form: BCNF

---

2. Customers Table

Functional Dependencies (FDs):

- customer_id → name, contact, address
- contact → customer_id, name, address *(if contact is unique)*

Checking Normal Forms:

UNF to 1NF: Already in 1NF (atomic values, no multi-valued attributes).
1NF to 2NF: No partial dependencies (all attributes depend on customer_id).
2NF to 3NF: No transitive dependencies.
3NF to BCNF: Every determinant is a candidate key.

Final Normal Form: BCNF

---

3. Medicines Table

Functional Dependencies (FDs):

- medicine_id → name, category
- medicine_id → price, stock_quantity, expiry_date
- name → category *(assuming each medicine belongs to only one category)*

Checking Normal Forms:

UNF to 1NF: Already in 1NF (atomic values, no repeating groups).
1NF to 2NF: Fails 2NF because expiry_date and price may depend on batches of medicine rather than just medicine_id.

Solution: Decomposing into Two Tables

Updated Medicines Table (Now in 2NF)

medicine_id	name	category
101	Paracetamol	Painkiller

New MedicineBatch Table (Now in 2NF)

batch_id	medicine_id	expiry_date
B1	101	2026-05-12

2NF to 3NF: No transitive dependencies.
3NF to BCNF: Every determinant is a candidate key.

Final Normal Form: BCNF

---

4. Suppliers Table

Functional Dependencies (FDs):

- **supplier_id → name, contact, address**
- **contact → supplier_id, name, address *(if contact is unique)***

Checking Normal Forms:

UNF to 1NF: No multi-valued attributes.
1NF to 2NF: No partial dependencies.
2NF to 3NF: No transitive dependencies.
3NF to BCNF: Every determinant is a candidate key.

Final Normal Form: BCNF

---

5. Orders Table

Functional Dependencies (FDs):

- **order_id → supplier_id, order_date, total_amount**
- **supplier_id → supplier_name, supplier_contact**

Checking Normal Forms:

UNF to 1NF: No multi-valued attributes.
1NF to 2NF: Fails 2NF because total_amount depends on ordered medicines, not order_id.

Solution: Decomposing into Two Tables

Updated Orders Table (Now in 2NF)

order_id	supplier_id	order_date
O1	S1	2025-03-10

New OrderDetails Table (Now in 2NF)

order_id	medicine_id	quantity
O1	101	50

2NF to 3NF: No transitive dependencies.
3NF to BCNF: Every determinant is a candidate key.

Final Normal Form: BCNF

---

6. InventoryLogs Table

Functional Dependencies (FDs):

- **log_id → medicine_id, change_type, quantity_changed, timestamp**

Checking Normal Forms:

Already in BCNF (no partial or transitive dependencies).

Final Normal Form: BCNF

---

7. Sales & SaleDetails Tables

Functional Dependencies (FDs):

- **sale_id → customer_id, sale_date, total_price**
- **sale_id, medicine_id → quantity, price**

Checking Normal Forms:

UNF to 1NF: No multi-valued attributes.
Fails 2NF: Price depends on medicine_id, not sale_id

Solution: Decomposing into Two Tables

Updated Sales Table (Now in 2NF)

sale_id	customer_id	sale_date
S101	1	2025-03-15

New SaleDetails Table (Now in 2NF)

sale_id	medicine_id	quantity	price
S101	101	2	25.50

Final Normal Form: BCNF

---

8. Prescriptions Table

Functional Dependencies (FDs):

- **prescription_id → customer_id, medicine_id, prescribed_by, date_issued**
- **customer_id → name, contact, address *(From Customers Table, but stored separately)***
- **medicine_id → name, category, price, expiry_date *(From Medicines Table, but stored separately)***

Why Is This in BCNF?

1. **Candidate Key: prescription_id uniquely determines all attributes.**
2. **No Partial Dependencies: Every attribute depends entirely on prescription_id.**
3. **No Transitive Dependencies:**
   - **prescription_id determines medicine_id, but medicine details are stored in a separate Medicines table.**
   - **customer_id determines customer details, but those are stored in Customers.**
   - **There is no case where a non-key attribute determines another non-key attribute.**

Conclusion: Already in BCNF.

---

9. PaymentTransactions Table

Functional Dependencies (FDs):

- **transaction_id → sale_id, payment_method, amount_paid, payment_date**
- **sale_id → customer_id, sale_date, total_price *(From Sales Table)***

Why Is This in BCNF?

1. **Candidate Key: transaction_id uniquely determines all attributes.**
2. **No Partial Dependencies: Every attribute depends fully on transaction_id.**
3. **No Transitive Dependencies:**
   - **sale_id determines customer_id (but this is stored separately in Sales table).**

- payment_method is atomic and does not determine other attributes.

Conclusion: Already in BCNF.

---

10. DiseaseOutbreakPredictions Table

Functional Dependencies (FDs):

- **prediction_id → predicted_disease, confidence_score, prediction_date**

Why Is This in BCNF?

1. **Candidate Key: prediction_id uniquely determines all attributes.**
2. **No Partial Dependencies: Every attribute depends fully on prediction_id.**
3. **No Transitive Dependencies:**
   - predicted_disease does not determine confidence_score.
   - confidence_score does not determine prediction_date.

Conclusion: Already in BCNF.

---

11. MedicineRecommendations Table

Functional Dependencies (FDs):

- **recommendation_id → prediction_id, medicine_id, recommendation_reason**
- **prediction_id → predicted_disease, confidence_score, prediction_date *(From DiseaseOutbreakPredictions Table)***
- **medicine_id → name, category, price, expiry_date *(From Medicines Table)***

Why Is This in BCNF?

1. **Candidate Key: recommendation_id uniquely determines all attributes.**
2. **No Partial Dependencies:**
   - medicine_id details are stored separately in Medicines table.
   - prediction_id details are stored separately in DiseaseOutbreakPredictions.
3. **No Transitive Dependencies:**
   - prediction_id determines predicted_disease but this is stored in a separate table.
   - medicine_id determines medicine details, but those are in Medicines.

Conclusion: Already in BCNF.

---

12. DemandAnalytics Table

Functional Dependencies (FDs):

- **analytics_id → medicine_id, total_sales, avg_daily_demand, last_updated**
- **medicine_id → name, category, price, expiry_date *(From Medicines Table)***

Why Is This in BCNF?

1. **Candidate Key: analytics_id uniquely determines all attributes.**
2. **No Partial Dependencies: Every attribute depends fully on analytics_id.**
3. **No Transitive Dependencies:**
   - **medicine_id determines medicine details, but those are stored separately in Medicines.**
   - **total_sales, avg_daily_demand, and last_updated all directly depend on analytics_id.**

Conclusion: Already in BCNF.

---

13. Reports Table

Functional Dependencies (FDs):

- **report_id → report_type, generated_date, report_content**

Why Is This in BCNF?

1. **Candidate Key: report_id uniquely determines all attributes.**
2. **No Partial Dependencies: Every attribute depends fully on report_id.**
3. **No Transitive Dependencies:**
   - **report_type does not determine generated_date.**
   - **generated_date does not determine report_content.**

Conclusion: Already in BCNF.